

VALLADOLID, Spain

WMO: 081410

Lat: 41.6408N

Lon: 4.7544W

Elev: 735

StdP: 92.80

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.4	-7.5	2.2	0.7	-5.7	2.5	1.5	9.7	7.0	8.2	6.3	1.4	20	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	34.6	18.2	32.9	17.8	31.2	17.3	19.5	30.9	18.8	30.0	18.1	28.8	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	12.4	21.6	15.2	11.8	21.0	14.4	11.2	20.5	58.4	31.1	56.1	30.1	53.9	28.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.1	5.2	-5.9	37.2	1.4	1.2	-7.0	38.1	-7.8	38.8	-8.6	39.4	-9.6	40.3		
			-6.4	21.5	1.4	1.8	-7.5	22.8	-8.3	23.8	-9.1	24.8	-10.2	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	4.6	5.9	8.9	11.1	15.0	19.9	22.4	22.1	18.3	13.8	7.9	4.9	(6)
		DBStd	7.15	3.03	2.97	3.18	3.43	3.72	3.85	2.99	2.97	3.20	3.24	3.25	3.19	
		HDD10.0	619	168	117	58	27	4	0	0	0	7	78	161		
		HDD18.3	2343	425	348	292	216	116	28	4	38	142	312	417		
		CDD10.0	1698	1	2	25	61	158	296	385	376	250	126	16	2	
		CDD18.3	382	0	0	0	1	12	74	132	122	38	3	0	0	
		CDH23.3	4693	0	0	1	22	196	939	1618	1438	436	42	0	0	
		CDH26.7	1988	0	0	0	1	48	401	760	647	127	4	0	0	
(14)	Wind	WSAvg	2.1	2.0	2.2	2.4	2.4	2.2	2.2	2.1	1.9	1.7	2.0	1.8		(14)
(15)	Precipitation	PrecAvg	445	46	34	34	44	48	32	15	14	31	50	52	44	(15)
(16)		PrecMax	702	183	103	130	116	163	95	67	68	107	160	166	157	(16)
(17)		PrecMin	190	0	0	0	3	8	0	0	0	3	0	2		(17)
(18)		PrecStd	111	37	26	30	27	31	24	17	16	27	36	38	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	13.1	17.2	21.7	25.7	30.7	36.2	36.4	36.2	32.7	27.0	18.7	13.8		(19)
(20)		0.4% MCWB	10.3	9.9	11.4	13.9	17.0	18.6	18.5	18.7	17.4	14.8	13.7	11.1		(20)
(21)		2% DB	11.5	14.6	19.6	23.4	27.9	33.4	34.5	34.2	30.0	24.3	16.0	12.1		(21)
(22)		2% MCWB	9.2	9.1	10.6	13.2	16.2	17.8	18.2	18.0	16.6	14.7	12.0	10.1		(22)
(23)		5% DB	10.4	12.7	17.7	20.9	25.5	31.0	32.9	32.5	27.8	21.8	14.2	10.9		(23)
(24)		5% MCWB	8.6	8.4	10.0	12.5	15.2	17.2	17.9	17.7	16.0	13.9	11.2	9.2		(24)
(25)		10% DB	9.2	11.1	15.5	18.4	23.0	28.8	31.1	30.5	25.9	19.7	12.9	9.6		(25)
(26)		10% MCWB	7.6	7.7	9.4	11.4	14.1	16.7	17.4	17.2	15.5	13.5	10.4	8.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.1	11.5	12.9	15.6	18.5	19.9	20.9	20.4	18.7	17.1	14.4	11.9		(27)
(28)		0.4% MCDB	12.2	14.8	18.8	23.4	28.2	30.6	32.4	31.5	28.4	22.5	17.1	12.9		(28)
(29)		2% WB	9.9	10.2	11.7	14.0	16.8	18.9	19.6	19.3	17.7	15.9	12.9	10.5		(29)
(30)		2% MCDB	11.1	13.2	16.9	21.4	25.8	30.4	31.1	31.0	27.3	21.0	15.1	11.8		(30)
(31)		5% WB	8.8	9.0	10.8	12.8	15.7	18.1	18.8	18.6	16.9	15.0	11.8	9.5		(31)
(32)		5% MCDB	10.0	11.8	16.1	19.4	24.0	28.8	30.1	29.8	25.5	19.7	13.7	10.7		(32)
(33)		10% WB	7.8	8.1	9.9	11.7	14.7	17.2	18.0	17.8	16.1	14.2	10.7	8.3		(33)
(34)		10% MCDB	9.0	10.8	14.7	17.6	21.7	27.1	28.8	28.4	24.0	18.4	12.5	9.6		(34)
(35)	Mean Daily Temperature Range	MDBR	6.6	9.1	10.9	11.2	12.2	14.1	15.2	14.9	13.0	10.3	7.5	6.5		(35)
(36)		5% DB MCDBR	7.8	12.0	15.3	15.9	16.3	17.4	17.8	17.5	16.4	14.2	9.5	7.0		(36)
(37)		5% WB MCWBR	5.5	7.2	7.8	7.4	6.4	5.6	5.5	5.7	5.9	6.3	6.0	5.1		(37)
(38)		5% WB MCDBR	6.3	10.2	12.6	14.0	14.7	15.9	16.3	15.8	13.9	10.9	8.0	6.1		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.283	0.294	0.324	0.336	0.347	0.352	0.337	0.342	0.330	0.318	0.294	0.280		(40)
(41)		taud	2.519	2.485	2.417	2.392	2.385	2.385	2.400	2.382	2.434	2.491	2.526	2.528		(41)
(42)		Ebn at Noon	881	923	926	934	926	919	929	916	907	880	859	854		(42)
(43)		Edh at Noon	71	86	104	114	118	118	115	114	101	84	70	65		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.79	2.83	4.07	5.24	6.39	7.24	7.54	6.58	5.06	3.30	2.08	1.63		(44)
(45)		RadStd	0.20	0.32	0.48	0.54	0.55	0.30	0.29	0.29	0.24	0.31	0.25	0.16		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	-0.40	-0.48	N/A	N/A	N/A	+61
	Regional (5 neighbors)	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	N/A	+55

Nomenclature: See separate page